

OSSP PRACTICAL WEEK-9

TASK-1: Implement a file copy utility using low-level Linux file I/O system calls such as open(), read(), write(), lseek() and close(). Compare its performance with standard library functions.

SOURCE CODE:

```
user@LAPX-0934-890: ~  
GNU nano 8.7.1 copy.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main(int argc, char *argv[])  
{  
    int src, dest;  
    char buffer[1024];  
    ssize_t n;  
  
    if (argc != 3)  
    {  
        printf("Usage: %s source destination\n", argv[0]);  
        return 1;  
    }  
  
    src = open(argv[1], O_RDONLY);  
    if (src < 0)  
    {  
        perror("Source file open failed");  
        return 1;  
    }  
  
    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);  
    if (dest < 0)  
    {  
        perror("Destination file open failed");  
        close(src);  
        return 1;  
    }  
  
    while ((n = read(src, buffer, sizeof(buffer))) > 0)  
    {  
        write(dest, buffer, n);  
    }  
  
    lseek(src, 0, SEEK_SET);  
  
    close(src);  
    close(dest);  
  
    printf("File copied successfully.\n");  
    return 0;  
}
```

OUTPUT:

```

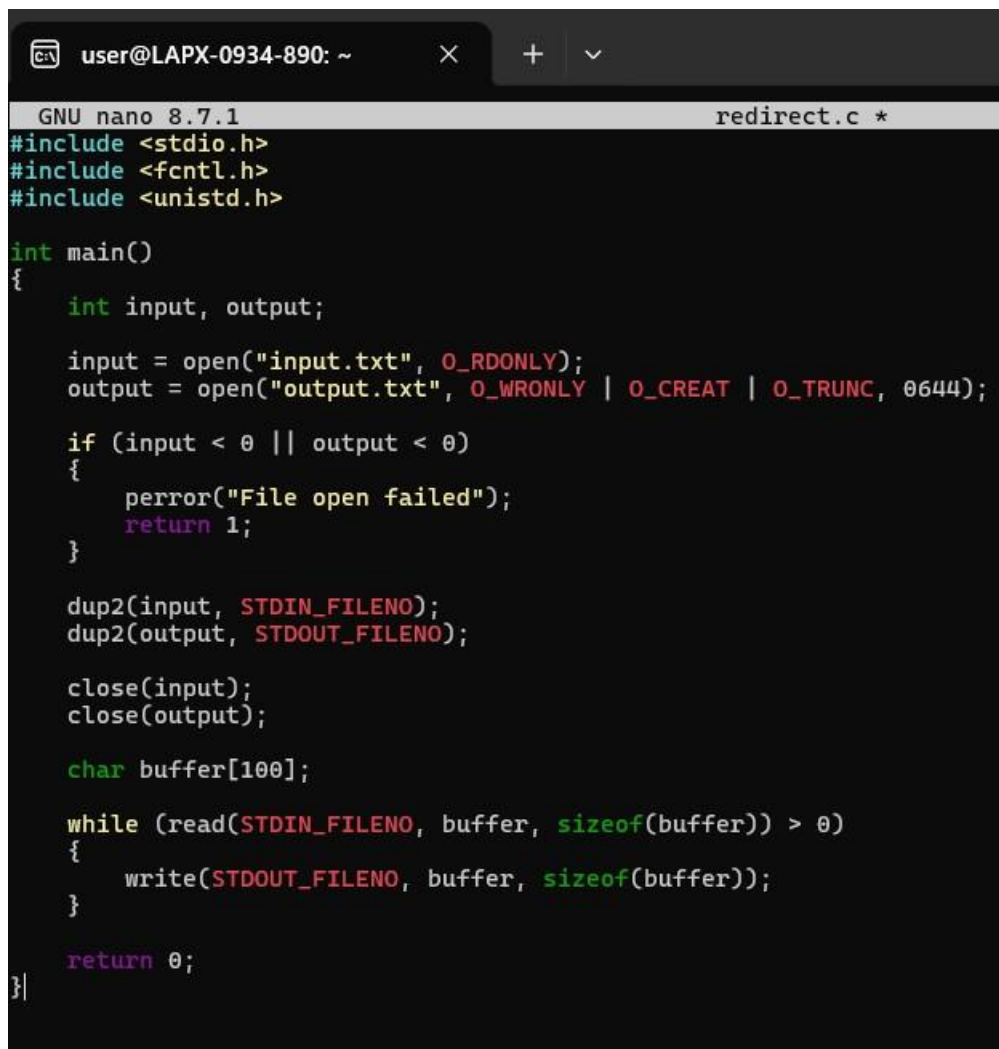
anuan@Anushka:~$ nano filecopy.c
anuan@Anushka:~$ gcc filecopy.c -o filecopy
anuan@Anushka:~$ echo "Hello Linux File I/O" > source.txt
anuan@Anushka:~$ ./filecopy source.txt destination.txt
File copied successfully.
Time taken: 0.000025 seconds
anuan@Anushka:~$ cat destination.txt
Hello Linux File I/O
anuan@Anushka:~$

```

OBSERAVTION: The source file was successfully copied to the destination file using low-level Linux file I/O system calls. The contents of both files were verified and found to be the same.

TASK 2 : Implement a file copy utility using low-level Linux file I/O system calls such as open(), read(), write(), lseek() and close(). Compare its performance with standard library functions.

SOURCE CODE :



```

GNU nano 8.7.1                                redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input, output;

    input = open("input.txt", O_RDONLY);
    output = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (input < 0 || output < 0)
    {
        perror("File open failed");
        return 1;
    }

    dup2(input, STDIN_FILENO);
    dup2(output, STDOUT_FILENO);

    close(input);
    close(output);

    char buffer[100];

    while (read(STDIN_FILENO, buffer, sizeof(buffer)) > 0)
    {
        write(STDOUT_FILENO, buffer, sizeof(buffer));
    }

    return 0;
}

```

OUTPUT :

```
anuan@Anushka:~$ nano filecopy2.c
anuan@Anushka:~$ gcc filecopy2.c -o filecopy2
anuan@Anushka:~$ echo "Hello Linux File I/O" > source.txt
anuan@Anushka:~$ ./filecopy2 source.txt destination.txt
File copied successfully.
Time taken: 0.000017 seconds
anuan@Anushka:~$ cat destination.txt
Hello Linux File I/O
anuan@Anushka:~$
```

OBSERVATION : The source file was successfully copied to the destination file using lowlevel Linux file I/O system calls. The contents of both files were verified and found to be the same.